IS229 WEB DESIGN

Week 06
JavaScript & DOM Interaction

Add behaviour to a page by connecting programming logic to user actions via the DOM.

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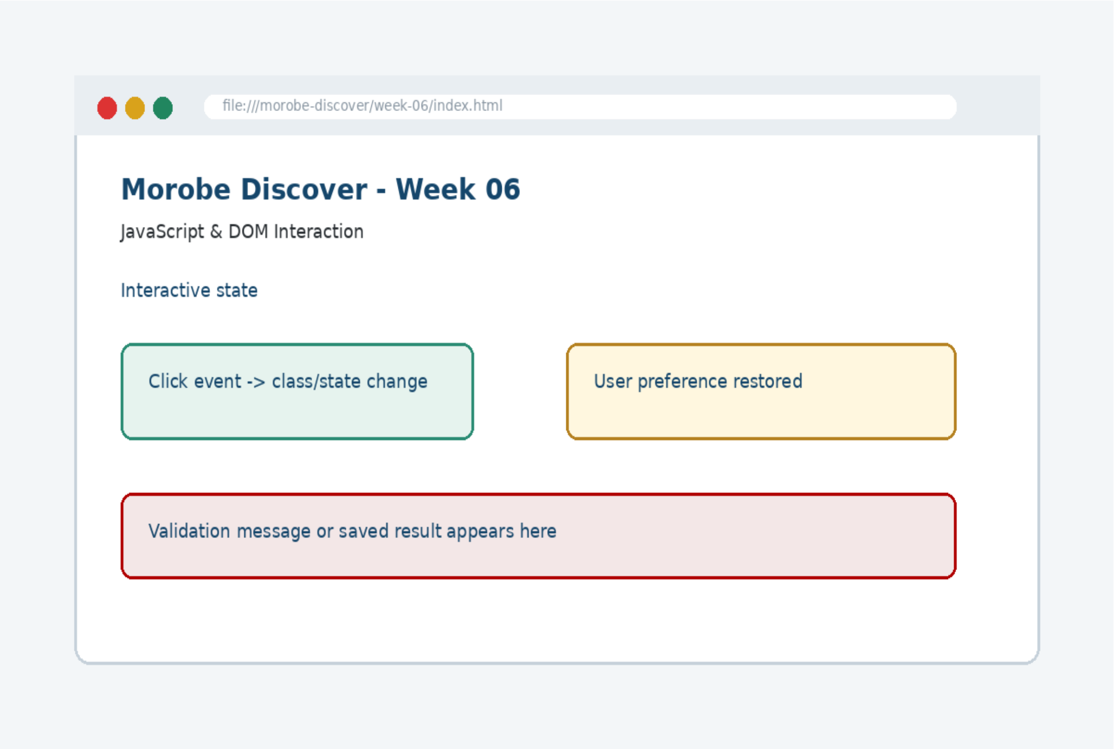
Learning outcomes

Add behaviour to a page by connecting programming logic to user actions via the DOM.
Explain the core programming/design concept for this week
Read and modify key code patterns
Test the weekly project increment
Prepare evidence for GitHub submission

Today's problem



Why it matters

Use JavaScript to listen for events, update state and change the interface deliberately.

Project before -> after

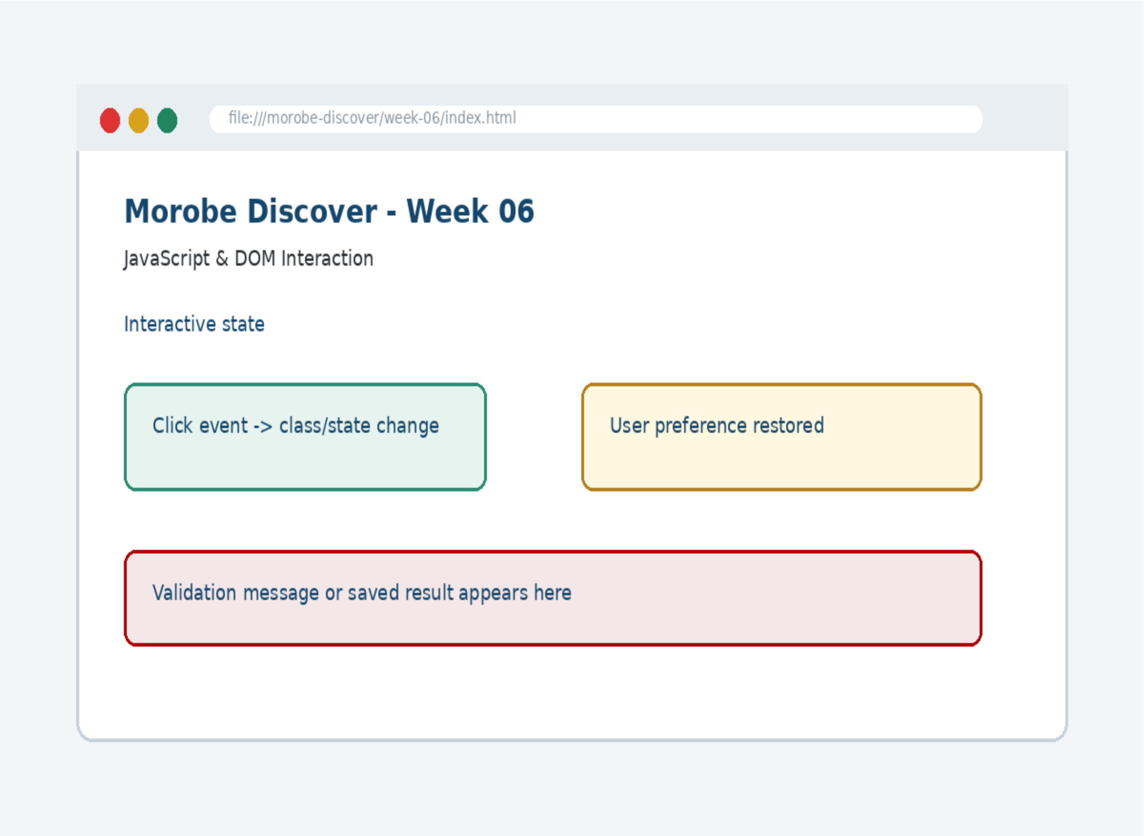
Before Week 06

Week 05 produced a responsive interface ready for interaction.

After Week 06

The site responds to user actions using JavaScript, DOM selection and class changes.

What we build this week



Add DOM hooks to the navigation
Write an event listener for the mobile menu
Toggle a state class instead of inline styles
Add a simple theme or FAQ interaction
Test interaction with click and keyboard focus

Core principle

Use JavaScript to listen for events, update state and change the interface deliberately.

Keep in mind

A good web developer can point to the code, the browser result, and the user reason behind the decision.

Key concepts

Concept 1

The DOM is the browser’s object model of the page

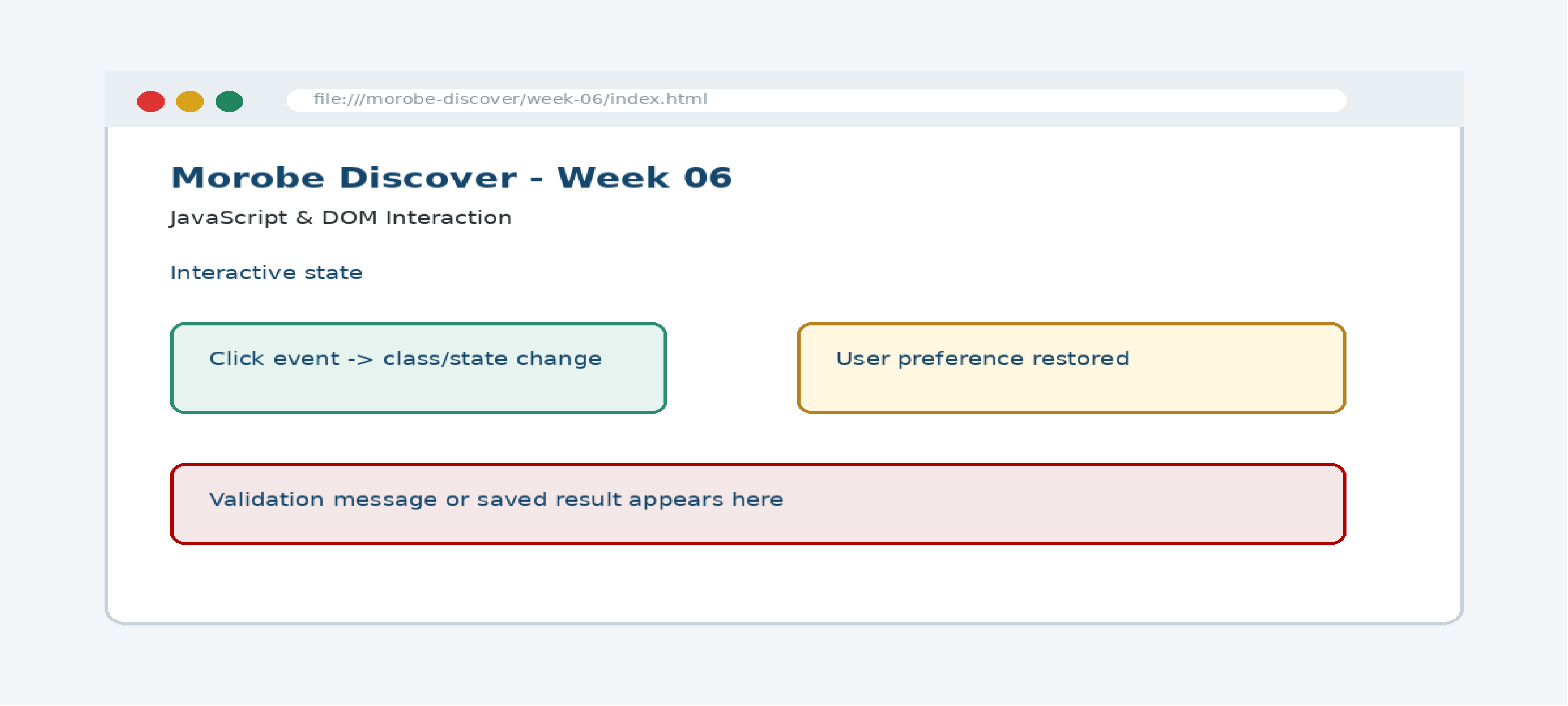
Concept 2

Event listeners connect actions to functions

Concept 3

Class toggling separates behaviour from style

Project anatomy



Key code pattern A

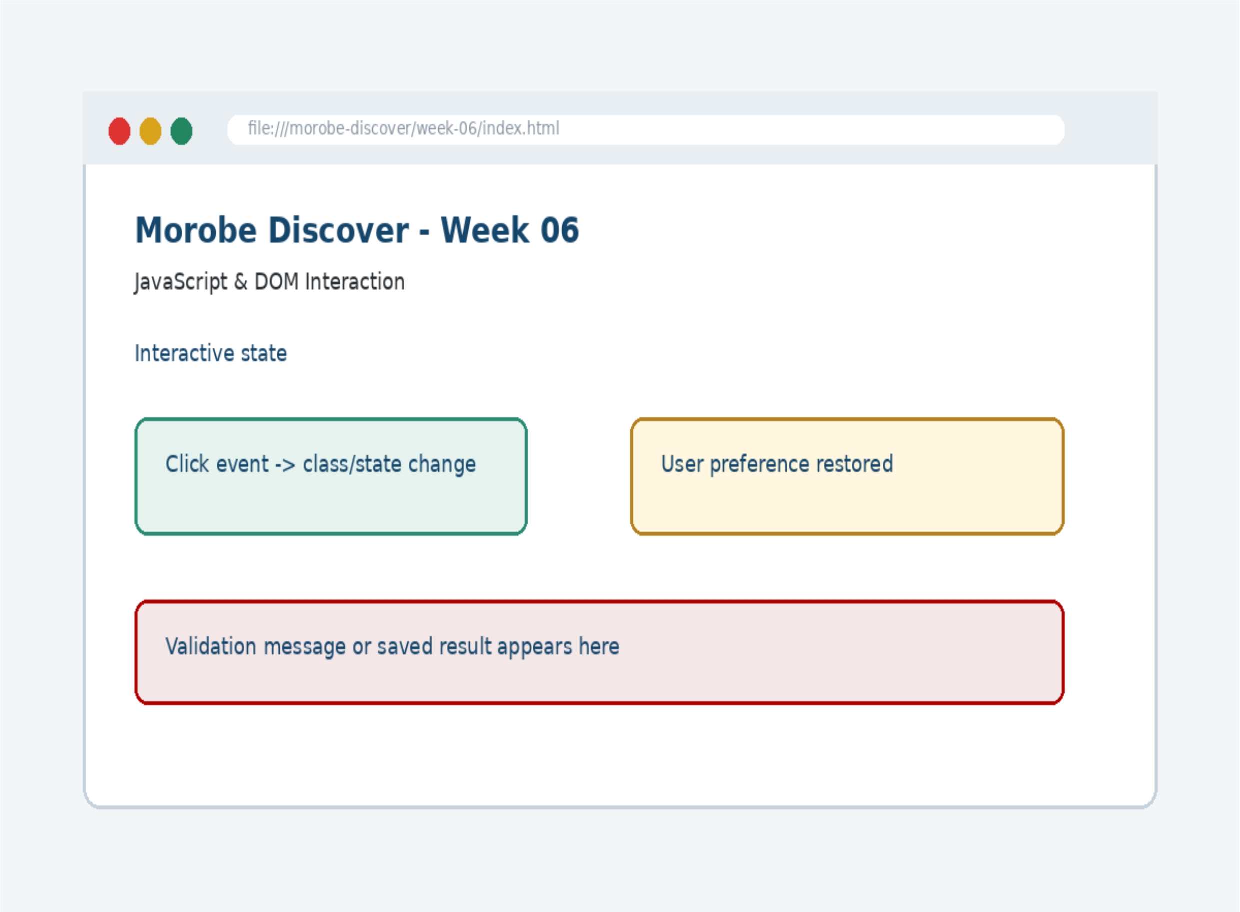
```
const menuButton =
document.querySelector(".menu-toggle");
const nav = document.querySelector(".site-nav");
menuButton.addEventListener("click", () => {
  nav.classList.toggle("is-open");
});
```

Explain it

Which file contains this code?
Which line creates the visible change?
What should appear in the browser?

Code -> visual result

```
const menuButton =
document.querySelector(".menu-toggle");
const nav =
document.querySelector(".site-nav");
menuButton.addEventListener("click", ()
=> {
  nav.classList.toggle("is-open");
});
```



What changed?

Project file changed for JavaScript & DOM Interaction
The browser output is now closer to the required weekly evidence
The change supports a user task or quality requirement
The change can be tested and committed

Key code pattern B

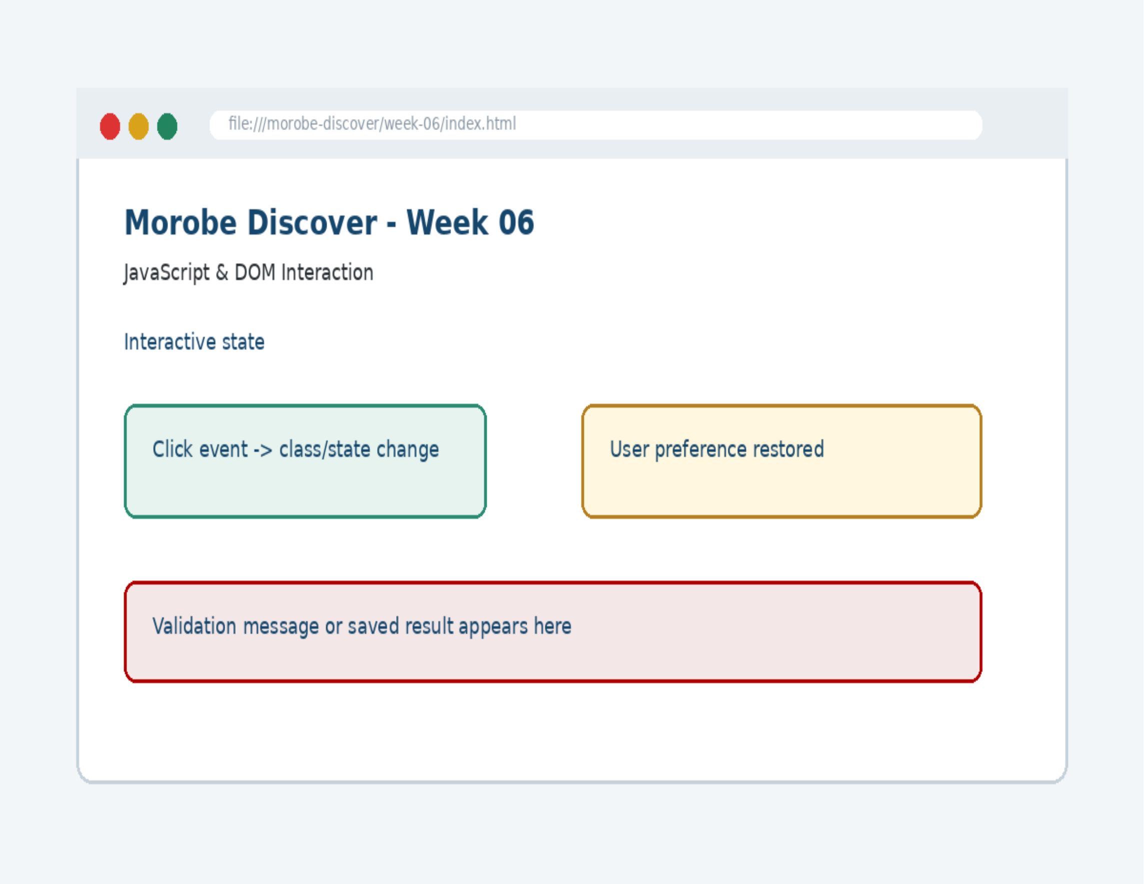
```
.site-nav.is-open {
  display: flex;
}
```

Why this pattern is needed

The second pattern usually applies the same idea to a different component, state, layout or data flow.

Apply it to Morobe Discover

Add DOM hooks to the navigation
Write an event listener for the mobile menu
Toggle a state class instead of inline styles
Add a simple theme or FAQ interaction



Compare the approaches

Weak approach

Memorise the definition
Copy code without understanding
No browser evidence
No test note

Professional approach

Explain the concept
Apply the correct code pattern
Inspect the result
Commit and describe the evidence

Common mistake

Mistake

The page appears unchanged, but the student assumes the feature is complete.

Correction

Check the correct file, refresh the browser, inspect DevTools, and verify the expected output.

Why does this fail?

```
// Broken checkpoint
// File changed, but the browser result does not
change.
// Possible causes:
// - wrong file
// - file not linked
// - syntax error
// - browser not refreshed
```

Class question

Which cause would you check first, and why?

Fix the problem

Confirm the browser is opening the correct file
Check links to CSS/JS/data files
Inspect the console for errors
Use a small working example
Test again before committing

Class activity

Activity - 7 minutes

Predict what happens when classList.toggle() runs twice.

Work in pairs
Point to the project evidence
Prepare one answer to share

Implementation steps 1-3

Add DOM hooks to the navigation
Write an event listener for the mobile menu
Toggle a state class instead of inline styles

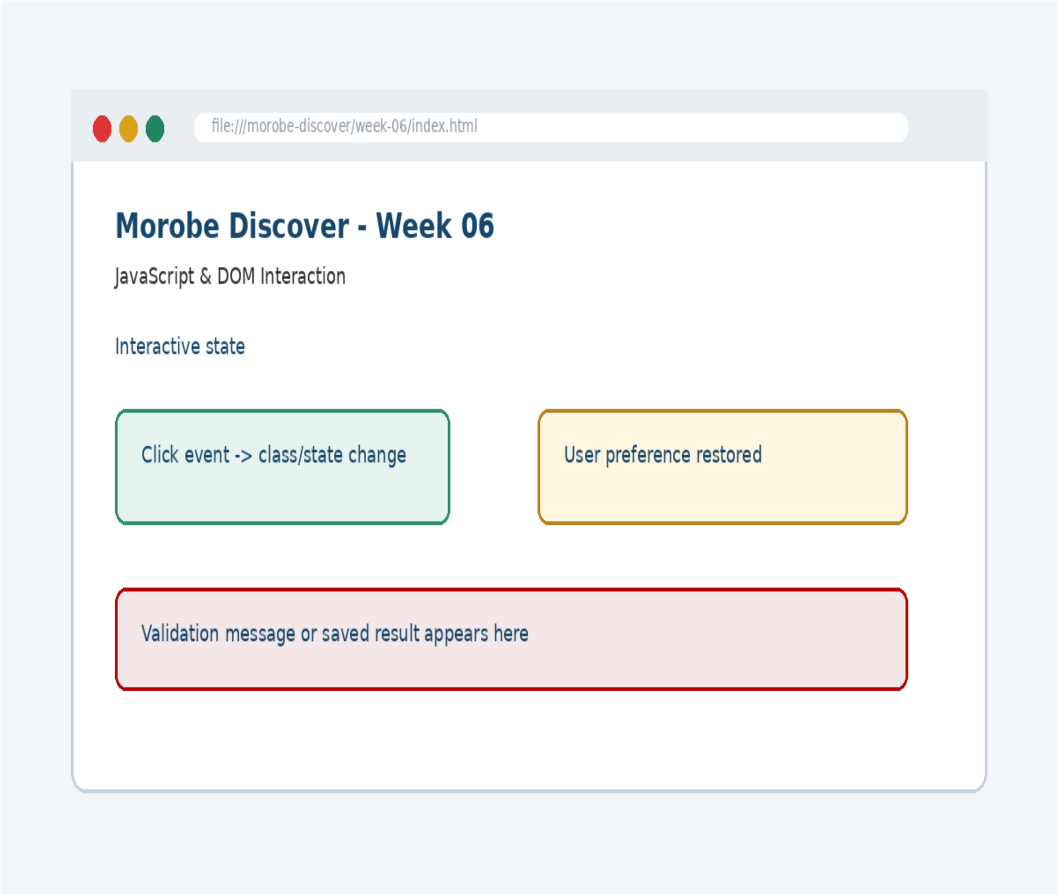
Implementation steps 4-5

Add a simple theme or FAQ interaction
Test interaction with click and keyboard focus

LIVE DEMO

Demo checklist

1. Open the current project
2. Locate the affected file
3. Add or modify the key code
4. Refresh the browser
5. Inspect the result
6. Explain the change



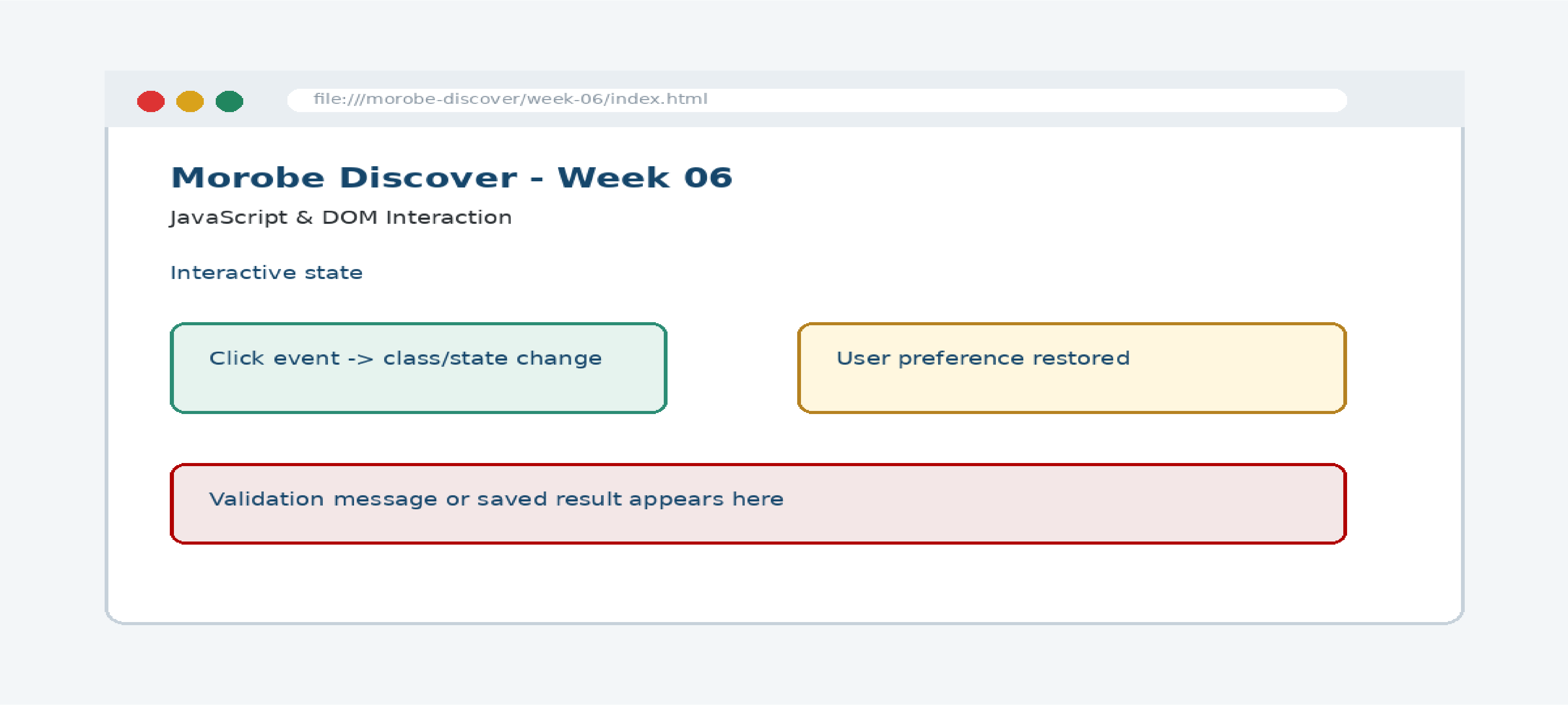
Predict the output

```
.site-nav.is-open {
  display: flex;
}
```

Prediction prompt

What should change on screen when this code is applied correctly?

Browser result



Testing the feature

Open the browser console
Click the menu several times
Check class changes in DevTools
Verify the site still works if JavaScript fails gracefully

Inspect with browser tools

Inspection path

Right click -> Inspect
Elements panel -> structure/classes
Console -> JavaScript errors
Network/Application -> data/storage when relevant

Evidence to save

Screenshot
Short test note
Changed files
Git commit message

Git evidence

```
git add .
git commit -m "Week 06 JavaScript & DOM Interaction
project increment"
```

Why it matters

Git history shows progression from Week 01 to final deployment. It is part of the assessment evidence.

Complete source code

```
Repository:
https://github.com/YOUR-USERNAME/is229-morobe-discover

Weekly folder:
/week-06/

Complete project:
/complete-project/
```

Reminder

Do not paste a whole file into a slide. Use the repository for complete code.

Mini challenge

Challenge

Modify one part of the Week 06 implementation and explain what changed in the browser.

State the file edited
State the line/section changed
Show the before and after result

What counts as evidence?

Technical evidence

Changed code files
Browser screenshot
Test note
Git commit

Understanding evidence

Code explanation
Why this method fits
Connection to previous week
Connection to next week

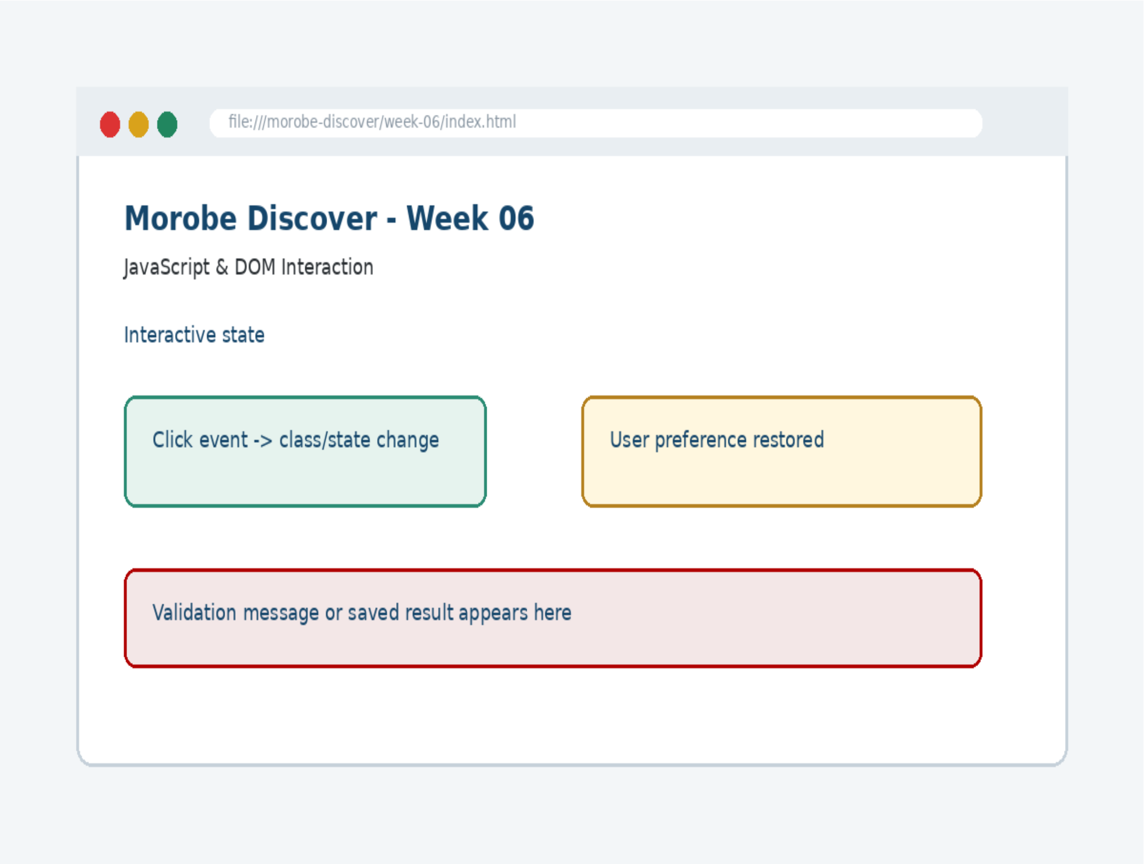
Project lab handoff

Add DOM hooks to the navigation
Write an event listener for the mobile menu
Toggle a state class instead of inline styles
Add a simple theme or FAQ interaction
Test interaction with click and keyboard focus

Definition of done

Week 06 feature is implemented
Project still preserves earlier week features
Browser result matches the expected output
Test note and screenshot are saved
Git commit message is meaningful

What we built today



The site responds to user actions using JavaScript, DOM selection and class changes.
Evidence: Event listeners, state changes, class toggling and interactive mobile menus
Technology: JavaScript DOM

Lecture summary

Use JavaScript to listen for events, update state and change the interface deliberately.
The DOM is the browser’s object model of the page
Event listeners connect actions to functions
Testing and Git are part of development, not extras

Prepare for next class

Finish this week’s lab before the next class
Read the next topic in advance
Bring the current project ready to extend
Be ready to explain one code decision

Exit question

How did Week 06 improve Morobe Discover, and what will the next week build on?